

NOBEL DANG

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EDUCATION

Clemson University | *Doctor of Philosophy in Computer Vision, Ph.D. (GPA: x / 4.0)* **SC, USA** | (2023 -Present)

- **Research Interest:** Novel View Synthesis, 3D Reconstruction, Localization, Foundational Models, Vision Language Models, Perception, Geometric Learning.
- **Coursework:** Computer Graphics, Linear and Nonlinear optimization.

New York University | *Master of Science in Computer Science (GPA: 3.97 / 4.0)* **NY, USA** | (2021 - 2023)

- **Coursework:** Computer Vision, Deep Learning, Machine Learning, Big Data, Cloud Computing, Information Visualization, Algorithms.

Guru Gobind Singh Indraprastha University | *B.Tech. in Computer Science & Engineering (GPA: 8.16 / 10)* **Delhi, India** | (2016 - 2020)

- **Coursework:** Artificial Intelligence, Machine Learning, Algorithms, Database Management Systems, Operating System, JAVA.

EXPERIENCE

VIPR, DEVCOM, CU (Research Scientist) **SC, USA** | (Aug 2023 – Present)

- Introducing hyperspectral-based foundation model with self-supervised learning.
- Developed **SOTA pose regression** based **visual localization** solution using 3D geometric foundation models that **improved the median pose error by 34%** and **rotation error by 37.5%**.

AI4CE, NYU (Research Assistant) **NY, USA** | (Jun 2022 – Aug 2023)

- Introduced and benchmarked a novel problem of **co-visibility reasoning (Co-Vision)** across family of deep learning models: VLMs, 3D Reconstruction, feature matching and contrastive.
- Developed the dataset for Co-Vision using **habitat-sim** and introduced a novel solution based on MV-DUST3R that **increased AUC performance** of vision models by **13%**.

Deep Learning, NYU (Teaching Assistant) **NY, USA** | (Jan 2023 – May 2023)

- Served as a Teaching Assistant in the Deep Learning course at NYU under Professor Chinmay Hegde.

Libsys Ltd (Software Developer Intern) **Gurugram, India** | (Jun 2019 – Aug 2019)

- Developed **hybrid application** for Library management system that uses RFID, using **Flutter**.

PUBLICATIONS

Co-Vision: Co-Visibility Reasoning on Sparse Image Sets of Indoor Scenes || [ICCV 2025](#) (**Outstanding Paper Award**)

- Developed Co-Vision benchmark, evaluating co-visibility reasoning of models across 1000+ indoor scenarios with sparse image set.
- Curated and open-sourced large-scale dataset using habitat-sim from iGibson and HM3D datasets by parallel processing.
- Introduced baselines for co-visibility reasoning; like traditional vision methods (**SIFT & RANSAC**), contrastive methods (**SimCLR with ViT**), place recognition (**with NetVLAD**), 3D reconstruction (**MV-DUST3R**) and multimodal methods (**GPT4-v and/or SigLIP**) with highest **AUC of 0.63**.
- Introduced Covis, a novel multi-view vision-based method, with **AUC of 0.57** and narrowed the gap to proprietary VLMs.

Kinematics Estimation of Carpal Bones || [MS Thesis, Proquest](#)

- Evaluated and estimated kinematics from volumetric 4D MR sequences of the carpal wrist bones by generating Dense Displacement vector fields in a novel way and detecting pathology in wrist using geometric learning from motion patterns.
- Performed 3D-segmentations and regression to get smooth trajectories of the carpal bones in SE(3) manifold.
- Created template of volumetric frame sequences of carpal bones using ANTs and ITK-SNAP and transferred the rigid segmentations from a high-resolution static image to the dynamic image sequences.
- Performed a novel quasi rigid image registration between the volume sequences that maintains the rigidity of carpal bones, but the rest of volume is deformable with average DICE score **~ 0.9**.

RESEARCH PROJECTS

Multiview3R || ***Currently in progress AutoAI@Clemson** ***Present**

- Developing end-to-end 3D reconstruction/NVS pipeline for outdoor environments.

Spatial-VPR || ***Currently in progress AutoAI@Clemson** ***Present**

- Introducing new methodology that solves VPR task by combing multi-view images to share information across them rather than individual image-based representation.
- Performed fundamental experiments with learnings from CroCo, CLIP, DIFT, MiDaS and DINOv2 to understand how they effect VPR task at scale.

Hyper-vision || ***Currently in progress AutoAI@Clemson** ***Present**

- Aligning hyperspectral and RGB modality for scene understanding.

SKILLS

- 3D Reconstruction, View Synthesis, Scene Understanding, Visual Localization, Large Language Models (and VLMs), Geometric Learning, Perception